

Protein Analysis Report

1. Protein Information

Protein Name: Insulin

UniProt Accession: P01308

Entry Name: INS_HUMAN

Organism: *Homo sapiens* (Human)

Gene Name: INS

Protein Length: 110 amino acids

Protein Status: Reviewed (UniProtKB/Swiss-Prot); Evidence at protein level

Function: Insulin is a peptide hormone that regulates blood glucose levels by promoting glucose uptake into cells. It also stimulates glycogen synthesis, protein synthesis, and lipid metabolism, thereby maintaining normal metabolic homeostasis.

2. FASTA Search

The amino acid sequence of human insulin (UniProt Accession: P01308) was retrieved from the UniProtKB database in FASTA format. The sequence was submitted to the NCBI BLASTP program and searched against the ClusteredNR protein database. The search identified numerous homologous insulin proteins from humans and other vertebrates with highly significant sequence similarity, confirming that the query sequence belongs to the insulin protein family.

3. FASTA Output

FASTA Header:

```
>sp|P01308|INS_HUMAN Insulin OS=Homo sapiens OX=9606 GN=INS PE=1 SV=1
```

Protein Sequence:

```
MALWMRLLPALLALLALWGPDPAAAFVNQHLCGSHLVEALYLVCGERGFFYTPKTRREAED  
LQVGQVELGGGPGAGSLQPLALEGSLQKRGIVEQCCTSIQSLYQLENYCN
```

Sequence Length: 110 amino acids

4. Interpretation

The protein sequence was analyzed using **NCBI BLASTP** against the **ClusteredNR** protein database. Several homologous insulin proteins were identified.

Top Alignment:

- **Organism:** *Pan troglodytes* (Chimpanzee)
- **Maximum Score:** 221 bits
- **Query Coverage:** 100%
- **E-value:** 1e-72
- **Sequence Identity:** 98.18% (108/110)
- **Positives:** 99% (109/110)
- **Gaps:** 0%

The **100% query coverage** indicates that the complete insulin sequence aligned with the matched protein. The **98.18% sequence identity** shows that the chimpanzee insulin protein is nearly identical to human insulin, reflecting a close evolutionary relationship. The extremely low **E-value (1e-72)** confirms that the alignment is highly significant and not due to random chance. The conserved insulin domains indicate that the protein retains its essential biological role in regulating glucose metabolism and maintaining metabolic homeostasis across closely related species.